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(a)i)	$\frac{C_{0.3}}{C_{0.1}} = \frac{d_{0.1}}{d_{0.3}}$ $C_{0.3} = \frac{0.1}{0.3}(0.089) = 0.030nF$																				
ii)	<table><caption>Approximate data points from the graph</caption><thead><tr><th>Separation d / mm</th><th>Capacitance C / nF (Purple Crosses)</th><th>Capacitance C / nF (Green Triangles)</th><th>Capacitance C / nF (Red Squares)</th></tr></thead><tbody><tr><td>0.1000</td><td>0.125</td><td>0.088</td><td>0.052</td></tr><tr><td>0.2000</td><td>0.062</td><td>0.045</td><td>0.028</td></tr><tr><td>0.3000</td><td>0.042</td><td>0.030</td><td>0.018</td></tr><tr><td>0.4000</td><td>0.032</td><td>0.022</td><td>0.014</td></tr></tbody></table>	Separation d / mm	Capacitance C / nF (Purple Crosses)	Capacitance C / nF (Green Triangles)	Capacitance C / nF (Red Squares)	0.1000	0.125	0.088	0.052	0.2000	0.062	0.045	0.028	0.3000	0.042	0.030	0.018	0.4000	0.032	0.022	0.014
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	B1 – plot correctly plotted B1 – smooth curve plotted passing through all points																				
iii)	$C = 0.082nF$																				
iv)	$C = \epsilon \frac{A}{d}$ $\epsilon = \frac{Cd}{A} = \frac{(0.082 \times 10^{-9})(0.15 \times 10^{-3})}{1.0 \times 10^{-4}}$																				
	$\epsilon = 1.23 \times 10^{-10} Fm^{-1}$																				
b)(i)	When time increases $e^{-t/RC}$ becomes very small, V_c will be very close to V_s																				
	This will result in an almost zero potential difference and zero current flow																				
(ii)	$I_R = \frac{V_R}{R}$ $= \frac{V_s}{R} e^{-t/RC}$																				
(c)(i)	Current flowing in the resistor determine how fast the charge is being discharged																				
	As the charges decreases, voltage decreases, current decreases and the rate of decrease in charge decreases.																				
(ii)	- Capacitor can be recharged - Capacitor can release their charge at a much faster rate																				

	- Capacitor do not lose their charge storage capacity over time
(d)(i)	$C = \frac{Q}{V}$ $Q = CV = (20 \times 10^{-6})(8) = 1.6 \times 10^{-4} C$
ii)	$V_c = V_s(1 - e^{-t/RC})$ $8 = 12(1 - e^{-t/(5)(20 \times 10^{-6})})$
	$t = 1.10 \times 10^{-4} s$
iii)	$F = \frac{1}{T}$ $= \frac{1}{1.10 \times 10^{-4}} = 9.09 kHz$
iv) 1.	The lower the resistance, the lower the amount of time the capacitor will take to reach 8.0 V
	Lowering the resistance will decrease the period and increase the frequency
2.	$2000 \leq \frac{1}{T}$ $T \leq 0.0005s$
	R = 10 and 20 Ω